Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

FOR FULL VIDEO SOLUTIONS
SCAN THE QR CODE
youtube.com/@hannahkettlemaths

Hannah Kettle MATHS TUTORIALS

Half Past Paper – Week 18

Higher Tier

You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser.
Tracing paper may be used.

Total Marks

Q	Topic	Marks	Sparx Code
2	Calculating with speed	3	U151
4a	Area and perimeter problems, Solving direct proportion word problems	4	U934, U721
4b	Area and perimeter problems, Solving direct proportion word problems	1	U934, U721
6a	Solving simultaneous equations graphically	1	U836
6b	Solving quadratic equations graphically	2	U601
8	Probabilities of mutually exclusive events, Expected results from repeated experiments	3	U683,U166
10a	Multiplying and dividing numbers in standard form	2	U264
10b	Multiplying and dividing numbers in standard form	2	U264
12a	Adding and subtracting algebraic fractions	3	U685
12b	Expanding triple brackets	3	U606
14	Circle theorems	4	U251,U130, U808
16a	Estimating areas under non-linear graphs	2	U882
16b	Estimating areas under non-linear graphs	1	U882
16c	Estimating gradients of non-linear graphs	1	U800
18	Using Pythagoras' theorem in 3D, Finding error intervals for calculations	4	U541,U587
20	Finding the LCM using prime factor decomposition, Index rules with positive indices	1	U250,U235
	Total	37	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 Andy cycles a distance of 30 km at an average speed of 24 km/h.
He then runs a distance of 12 km at an average speed of 8 km/h.

Work out the total time Andy takes.
Give your answer in hours and minutes.

..... hours ..... minutes

(Total for Question 2 is 3 marks)

4 Maisie knows that she needs 3 kg of grass seed to make a rectangular lawn 5 m by 9 m.

Grass seed is sold in 2 kg boxes.

Maisie wants to make a rectangular lawn 10 m by 14 m.
She has 5 boxes of grass seed.

(*a*) Has Maisie got enough grass seed to make a lawn 10 m by 14 m?
You must show all your working.

(4)

Maisie opens the 5 boxes of grass seed.

She finds that 4 of the boxes contain 2 kg of grass seed.
The other box contains 1 kg of grass seed.

(*b*) Does this affect whether Maisie has enough grass seed to make her lawn?
Give a reason for your answer.

.....

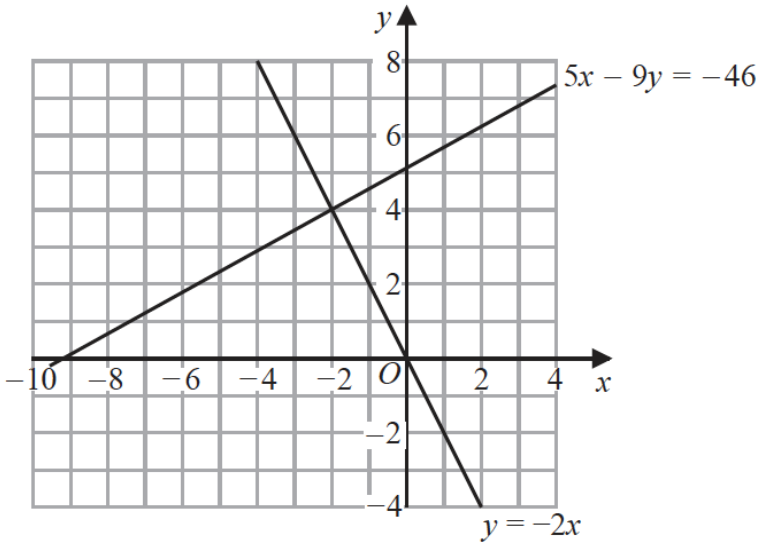
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(1)

(Total for Question 4 is 5 marks)

6



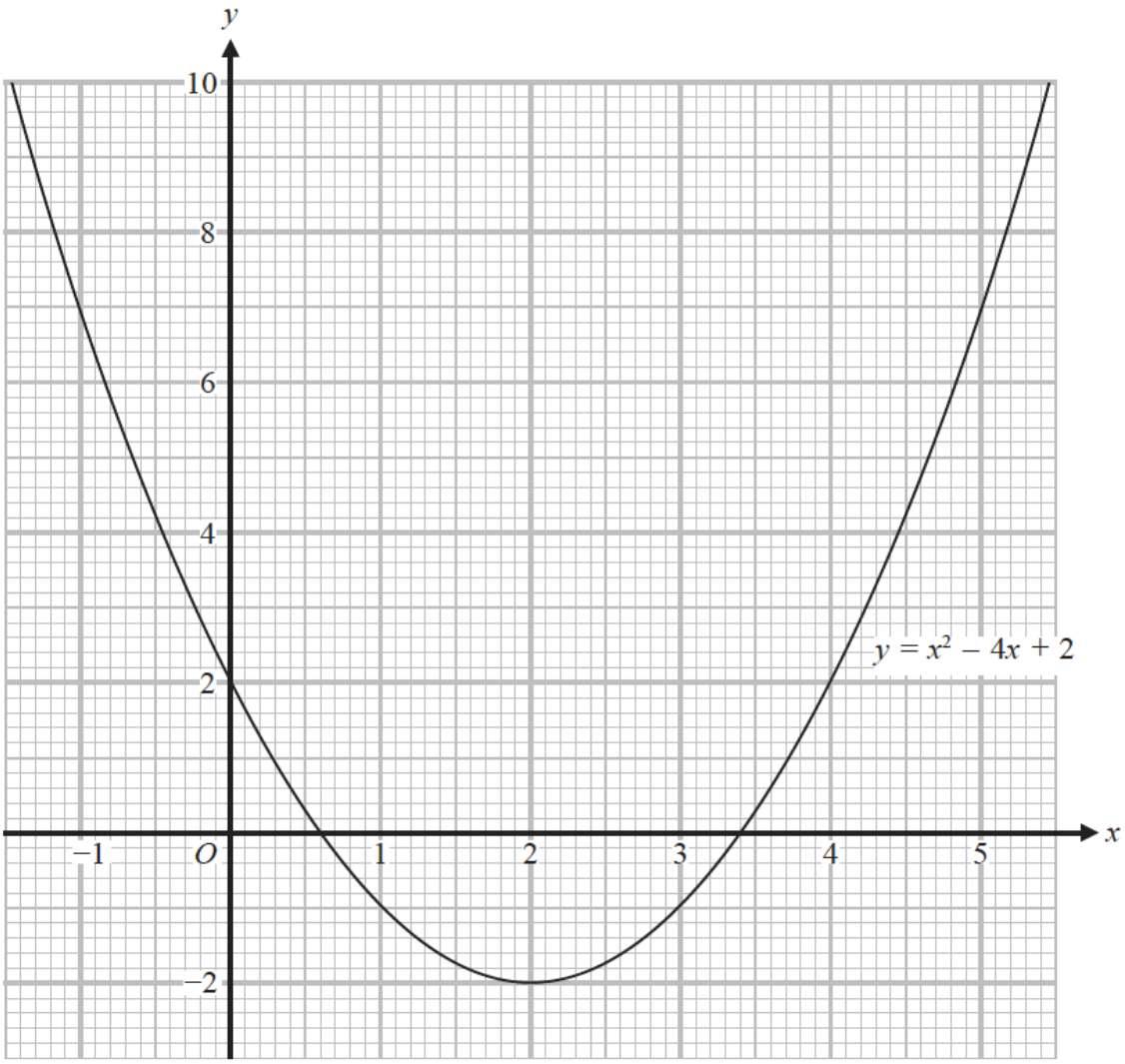
(*a*) Use these graphs to solve the simultaneous equations

$$5x - 9y = -46$$
$$y = -2x$$

$x =$ .....

$y =$ .....

(1)



(*b*) Use this graph to find estimates for the solutions of the quadratic equation $x^2 - 4x + 2 = 0$

.....

(2)

(Total for Question 6 is 3 marks)

8 There are some counters in a bag.
The counters are blue or green or red or yellow.

The table shows the probabilities that a counter taken at random from the bag will be blue or will be green.

Colour	blue	green	red	yellow
Probability	0.32	0.20		

The probability that a counter taken at random from the bag will be red is five times the probability that the counter will be yellow.

There are 300 counters in the bag.

Work out the number of yellow counters in the bag.

.....

(Total for Question 8 is 3 marks)

10 A person's heart beats approximately 10^5 times each day.
A person lives for approximately 81 years.

(*a*) Work out an estimate for the number of times a person's heart beats in their lifetime.
Give your answer in standard form correct to 2 significant figures.

.....

(2)

2×10^{12} red blood cells have a total mass of 90 grams.

(*b*) Work out the average mass of 1 red blood cell.
Give your answer in standard form.

..... grams

(2)

(Total for Question 10 is 4 marks)

12 (*a*) Express $\frac{x}{x+2}+\frac{2x}{x-4}$ as a single fraction in its simplest form.

.....

(3)

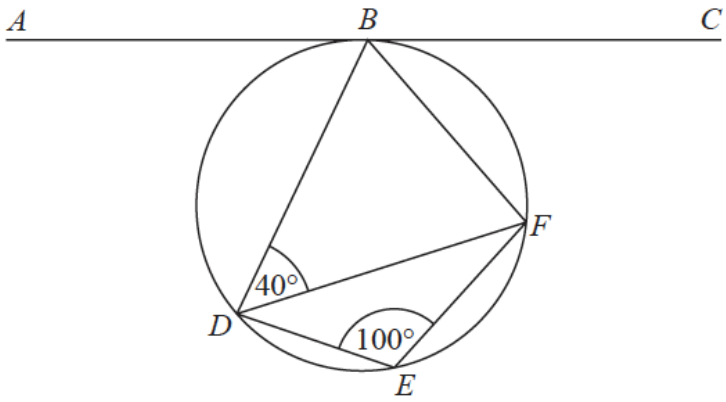
(*b*) Expand and simplify $(x - 3)(2x + 3)(4x + 5)$

.....

(3)

(Total for Question 12 is 6 marks)

14

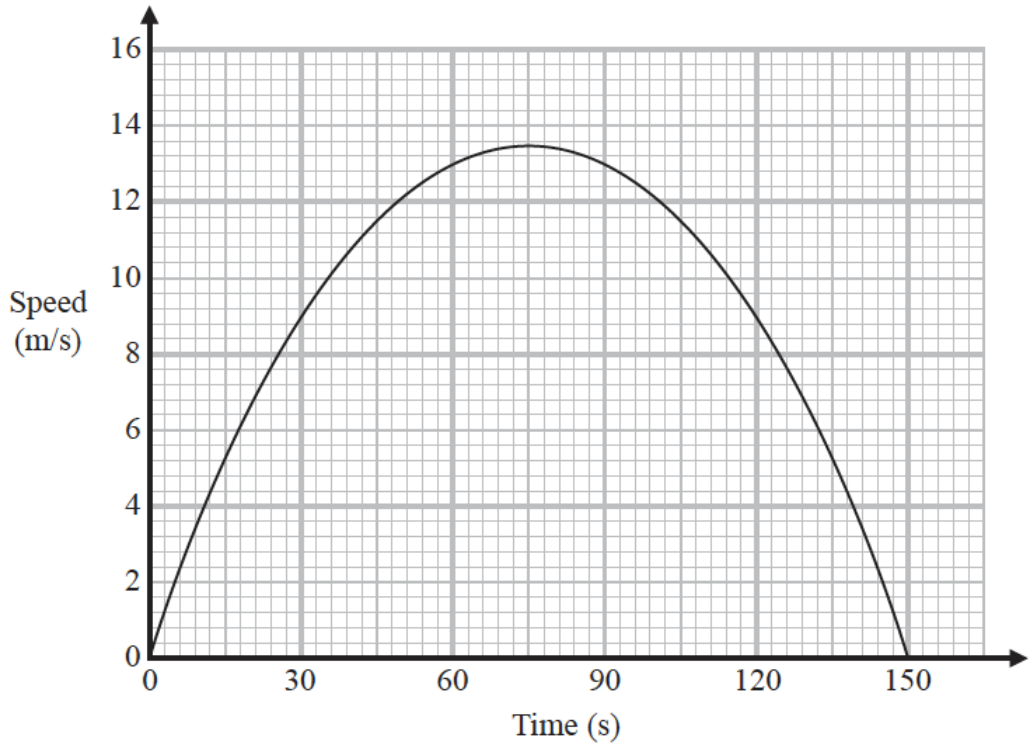


Points B, D, E and F lie on a circle.
ABC is the tangent to the circle at B.

Find the size of angle ABD.
You must give a reason for each stage of your working.

(Total for Question 14 is 4 marks)

16 Here is a speed-time graph for a car.



(*a*) Work out an estimate for the distance the car travelled in the first 30 seconds.

..... m

(2)

(*b*) Is your answer to part (*a*) an underestimate or an overestimate of the actual distance the car travelled in the first 30 seconds?
Give a reason for your answer.

.....

.....

.....

(1)

Julian used the graph to answer this question.

Work out an estimate for the acceleration of the car at time 60 seconds.

Here is Julian's working.

$$\text{acceleration} = \text{speed} \div \text{time}$$
$$= 13 \div 60$$
$$= 0.21\dot{6} \text{ m/s}^2$$

Julian's method does not give a good estimate of the acceleration at time 60 seconds.

(*c*) Explain why.

.....

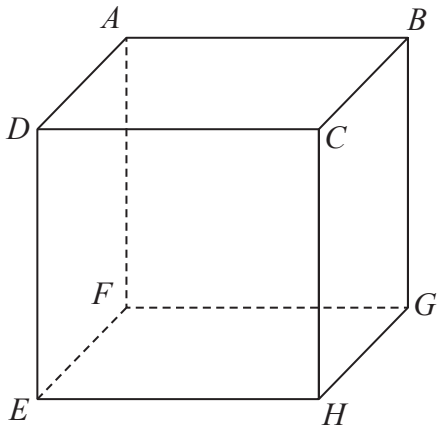
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.....

(1)

(Total for Question 16 is 4 marks)

18 The diagram shows a cube.



AH = 11.3 cm correct to the nearest mm.

Calculate the lower bound for the length of an edge of the cube.
You must show all your working.

..... cm

(Total for Question 18 is 4 marks)

20 Here is a list of five numbers.

98^{53} 98^{64} 98^{73} 98^{88} 98^{91}

Find the lowest common multiple of these five numbers.

.....

(Total for Question 20 is 1 mark)

TOTAL FOR PAPER IS 37 MARKS